2023-Ph-H1-Q12

Section: Electricity

Topic: Semiconductors, Sources, Internal Resistance

A 6 V supply with internal resistance of 1 Ω is connected to a 5 Ω resistor. What is the terminal potential difference?

Answer: D

(5 V)

Guidance for Students:

Calculate current first:
$I = \frac{\mathcal{E}}{R+r} = \frac{6}{5+1} = 1\,\mathrm{A}$

Then:
$V_{\text{terminal}} = \mathcal{E} - Ir = 6 - 1 = 5\,\mathrm{V}$

Revision Tips:

- Internal resistance reduces terminal voltage.
- Formula: $V = \mathcal{E} - Ir$
- Don’t forget: total resistance = external + internal.